Lab 3: Propositional Logic & Model Checking

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Task 1 - Understand the Code

1.1 harry.py: The knowledge base is true in only one of the eight possible assignments: rain = True, hagrid = False, dumbledore = True.

rain	hagrid	dumbledore	KB
False	False	False	False
False	False	True	False
False	True	False	False
False	True	True	False
True	False	False	False
True	False	True	True
True	True	False	False
True	True	True	False

Because rain is true in every model satisfying the knowledge base, model_check(knowledge, rain) returns True.

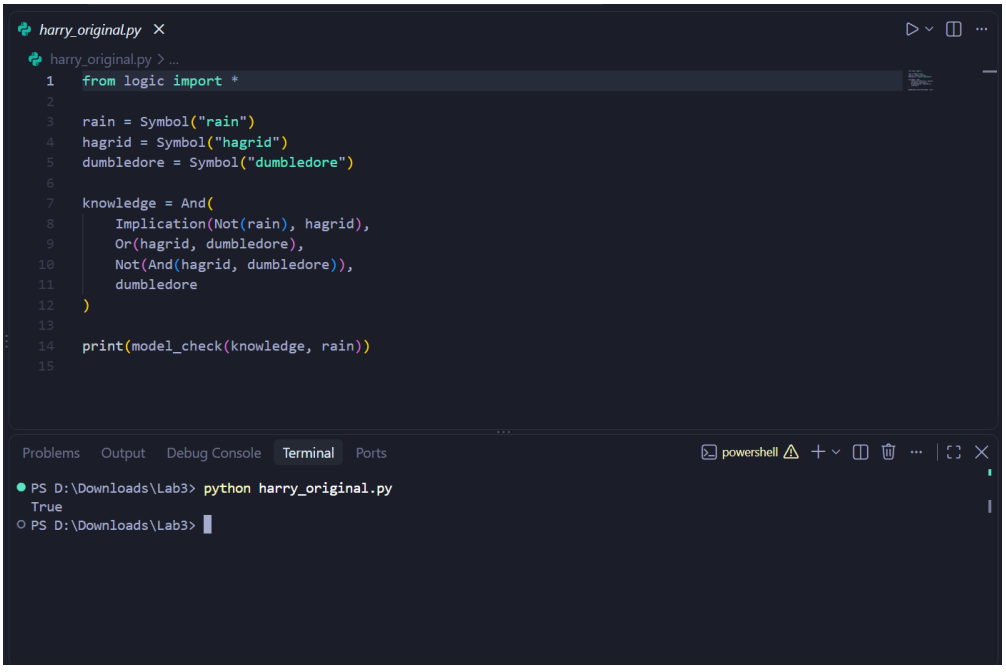


Figure 1. Original harry.py: True.

Task 1 - clue.py

ColMustard and ProfPlum are excluded, so MsScarlet is the suspect. Kitchen and ballroom are excluded, leaving library. Revolver is excluded, and the unknown-card constraint eliminates wrench. The remaining weapon is knife.

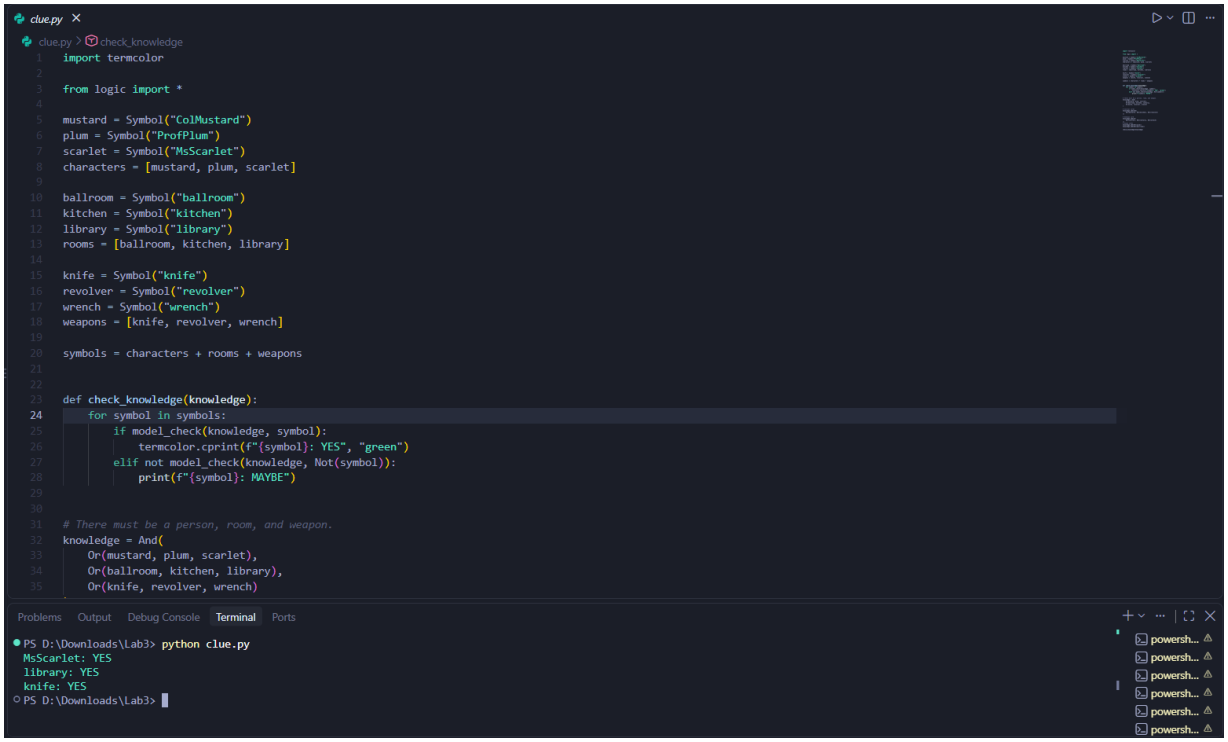


Figure 2. clue.py: MsScarlet, library, and knife.

Task 2 - Modify harry.py

I removed the final dumbledore fact from the knowledge base.

Prediction: False. Actual output: False.

Rain is no longer guaranteed. For example, rain = False, hagrid = True, and dumbledore = False satisfies the remaining rules. Therefore, the knowledge base does not entail rain.

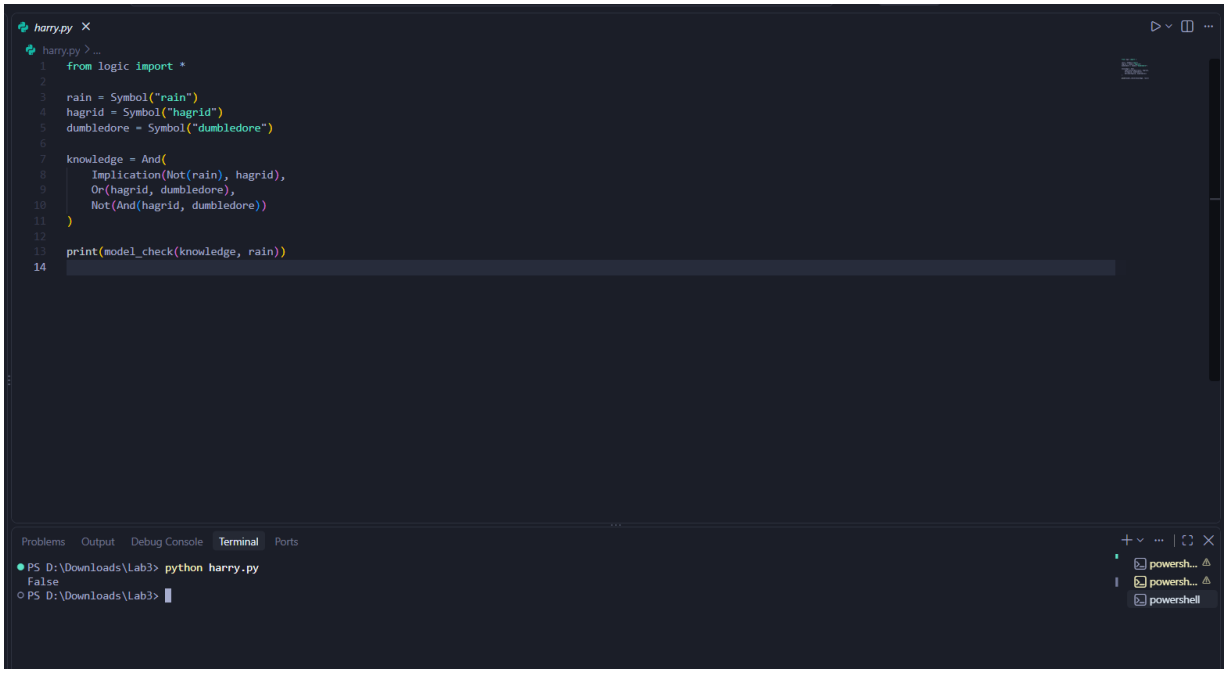


Figure 3. Modified harry.py: False.

Task 2 - Remove the Minerva Clue

I removed MinervaGryffindor from puzzle.py.

Prediction: No certain house assignments. Actual output: No assignments printed.

Eight valid house assignments remain, and no person has the same house in every valid assignment.

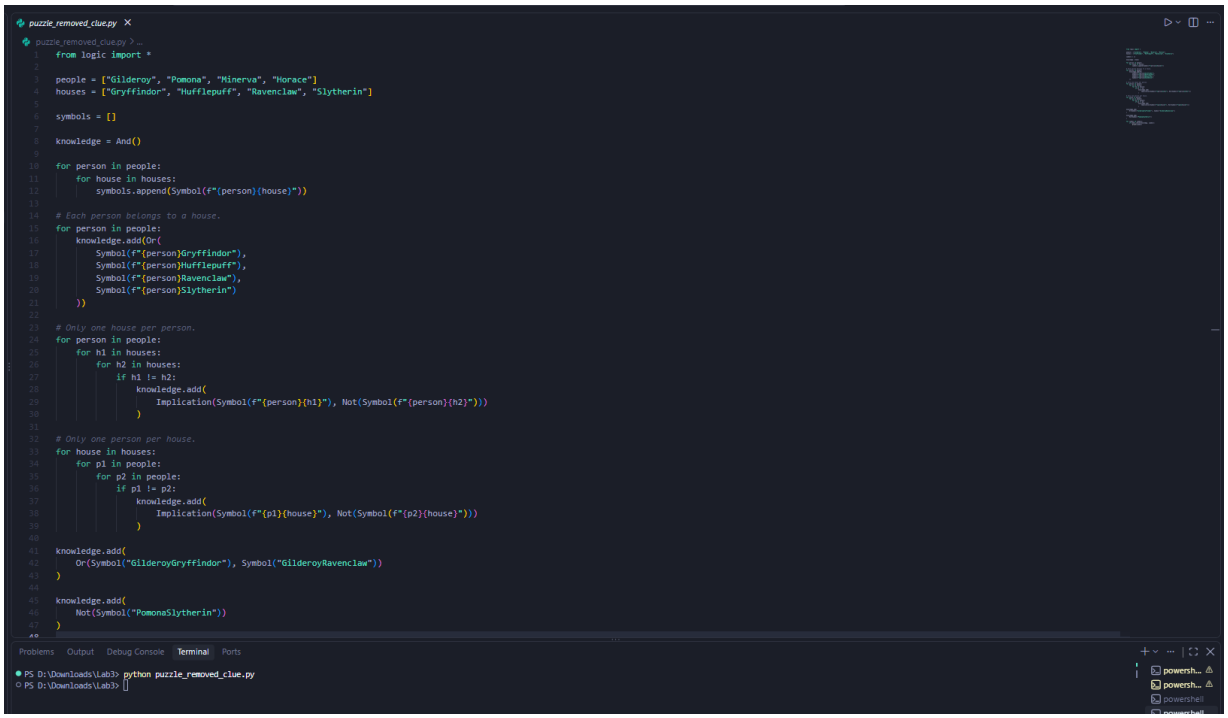


Figure 4. puzzle.py with the Minerva clue removed.

Task 2 - Add a New Clue

I added a compound clue using And: Gilderoy is in Ravenclaw, Pomona is in Hufflepuff, and Horace is in Slytherin. This leaves Gryffindor for Minerva and gives one unique solution.

Prediction: One complete assignment. Actual output: All four assignments.

Person	House
Gilderoy	Ravenclaw
Pomona	Hufflepuff
Minerva	Gryffindor
Horace	Slytherin

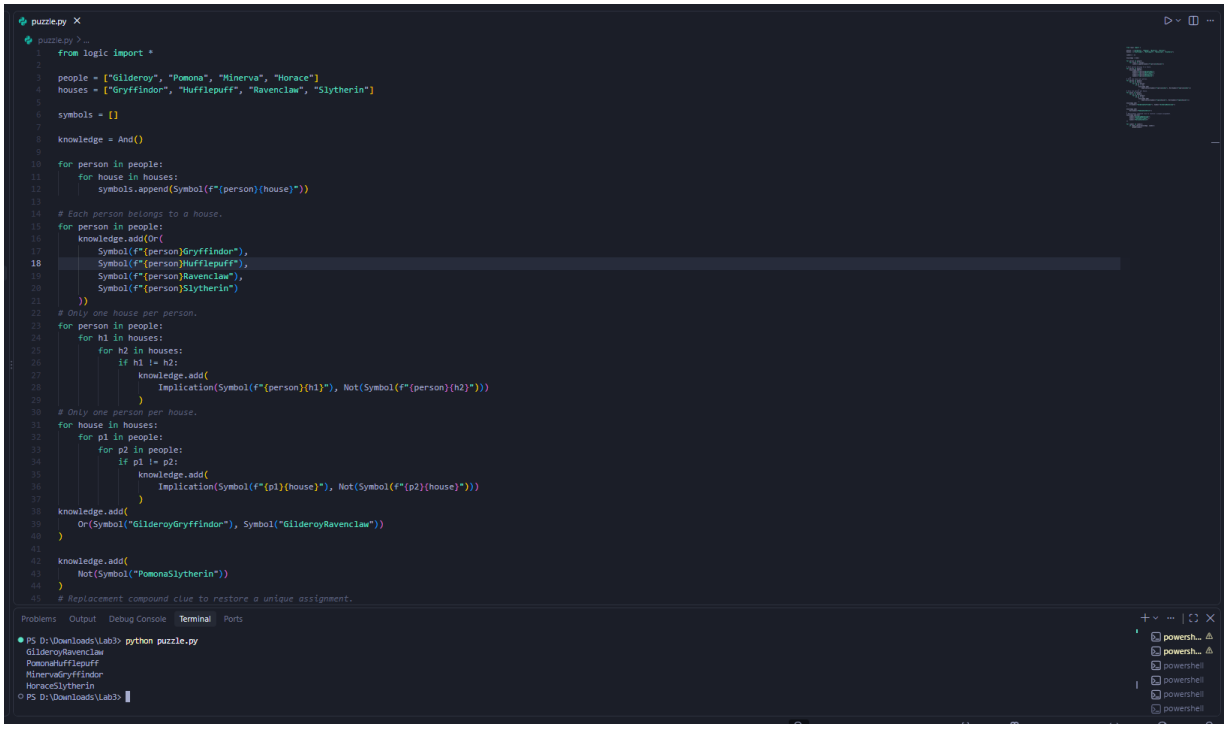


Figure 5. puzzle.py after adding the compound clue.

Task 3 - Knights and Knaves

Each person must be a knight or a knave, but not both. A knight tells the truth, while a knave lies.

Puzzle	Result
0	A: Knave
1	A: Knave; B: Knight
2	A: Knave; B: Knight
3 (Bonus)	A: Knight; B: Knave; C: Knight

Puzzle 0: A cannot be both types, so A lies. Puzzle 1: A cannot truthfully say both are knaves; B is therefore a knight. Puzzle 2: A lies about being the same kind, while B truthfully says they are different.

Bonus Puzzle 3: A said "I am a knight." B lies about what A said, so B is a knave. B also lies about C, making C a knight. C confirms that A is a knight.

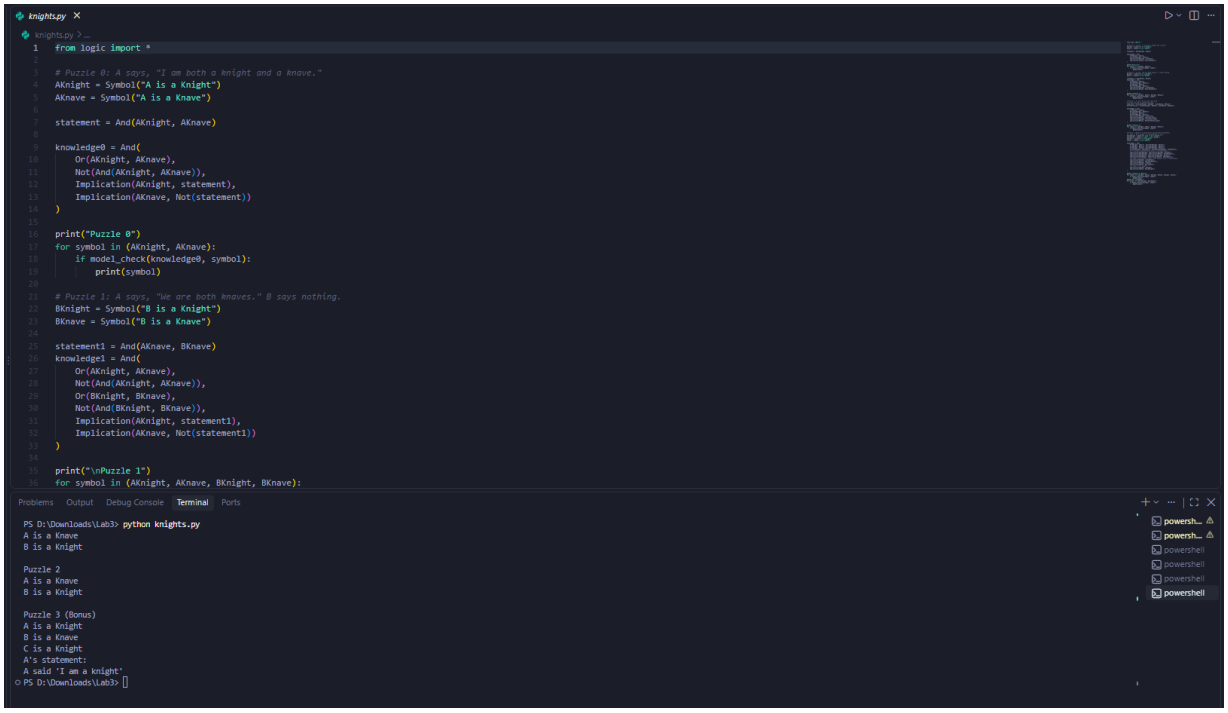


Figure 6. knights.py results for Puzzles 0-3 (saved output).

Conclusion: Model checking confirmed the expected results. Changing the knowledge base changed what could be logically proved.